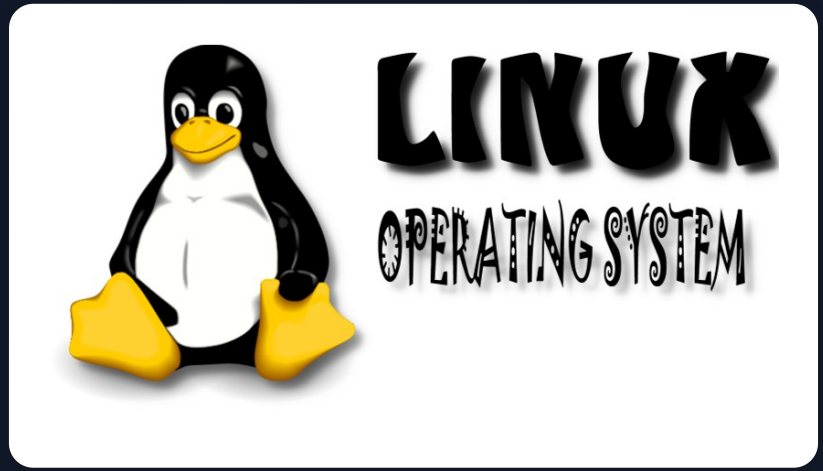
AN INTRODUCTION

Linux Operating System



The free, open-source operating system that powers servers, phones and supercomputers.

AGENDA

Contents

01 Operating System & Features
02 Definition & Naming of Linux
03 History of Linux
04 Components of Linux
05 Basic Features of Linux
06 Architecture of Linux
07 About the Kernel
08 User-Interface View of Linux
09 CLI / GUI View of Linux
10 Distributions of Linux
11 Hardware Requirements
12 Software Applications
13 Editors of Linux
14 Linux vs Other Operating Systems
15 Important Commands
16 Merits / Demerits of Linux
17 Use of Linux in Various Fields
18 Commercial Use of Linux
19 Conclusion

FUNDAMENTALS

Operating System



WHAT IS IT?

An operating system is software that communicates with the hardware and allows other programs to run.

Features of an operating system



Task Scheduling



Memory Management



Network Communication Handling



Data and User Security

FUNDAMENTALS

Definition & Naming of Linux

DEFINITION

Linux is a freely distributable, cross-platform, open-source operating system based on Unix. It can be installed on PCs, laptops, notebooks, mobile and tablet devices, video game consoles, servers, supercomputers and more.



Developed by Linus Torvalds

WHY THE NAME “LINUX”?

Torvalds wanted to call his invention “Freax” (Free + Freak + x), an allusion to Unix. His partner, Mr. A. L. Torvalds, did not think it was a good name, so they finally named the project “Linux”.

FUNDAMENTALS

History of Linux

1 — 2 — 3 — 4

1969

UNIX

Thompson & Ritchie, AT&T Bell Labs.

Vendors

Commercial Vendors

Sun, HP, IBM, SGI, DEC.

1984

GNU

Richard Stallman and the FSF.

GPL

Open Source

Software shared under the GPL licence.

FUNDAMENTALS

Components of the Linux System



Kernel

The core part of Linux. It is responsible for all major activities of the operating system.



System Library

Special functions or programs that applications and system utilities use to access the kernel's features.



System Utility

Programs responsible for specialized, individual-level tasks.

FUNDAMENTALS

Basic Features of Linux



Portable

Software works the same way on different types of hardware.



Open Source

Source code is freely available and developed by the community.



Multiprogramming

Multiple applications can run at the same time.

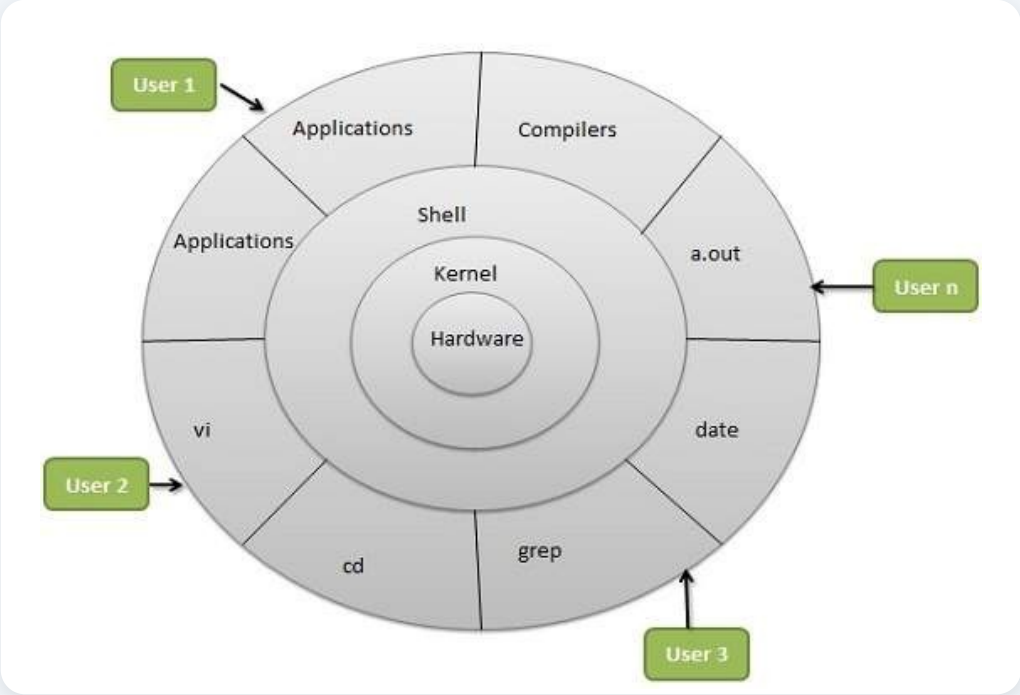


Security

Password protection, controlled access to files and data encryption.

ARCHITECTURE

Architecture of Linux



LAYERS AT A GLANCE

1. Hardware
2. Kernel
3. Shell
4. Utilities

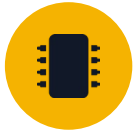
ARCHITECTURE

The Layers of Linux

Hardware layer	All peripheral devices (RAM, HDD, CPU etc.).
Kernel	Core component of the OS. Interacts directly with hardware and provides low-level services to the layers above.
Shell	Takes commands from the user and executes the kernel's functions.
Utilities	Programs that give users most of the functionality of an operating system.

ARCHITECTURE

About the Kernel



Talks to the hardware

It interacts with the actual hardware in machine language.



The monarch of the system

It has overall control of everything.

KEY FUNCTIONS

- File management
- File-system to hardware data transfer
- Memory management
- Scheduling of programs in memory
- Handling interrupts

INTERFACE

User Interface of Linux

The user interface is either a command line interface (CLI), a graphical user interface (GUI), or controls associated with hardware.

CLI **GUI**



CLI shells are text-based user interfaces, which use text for both input and output.



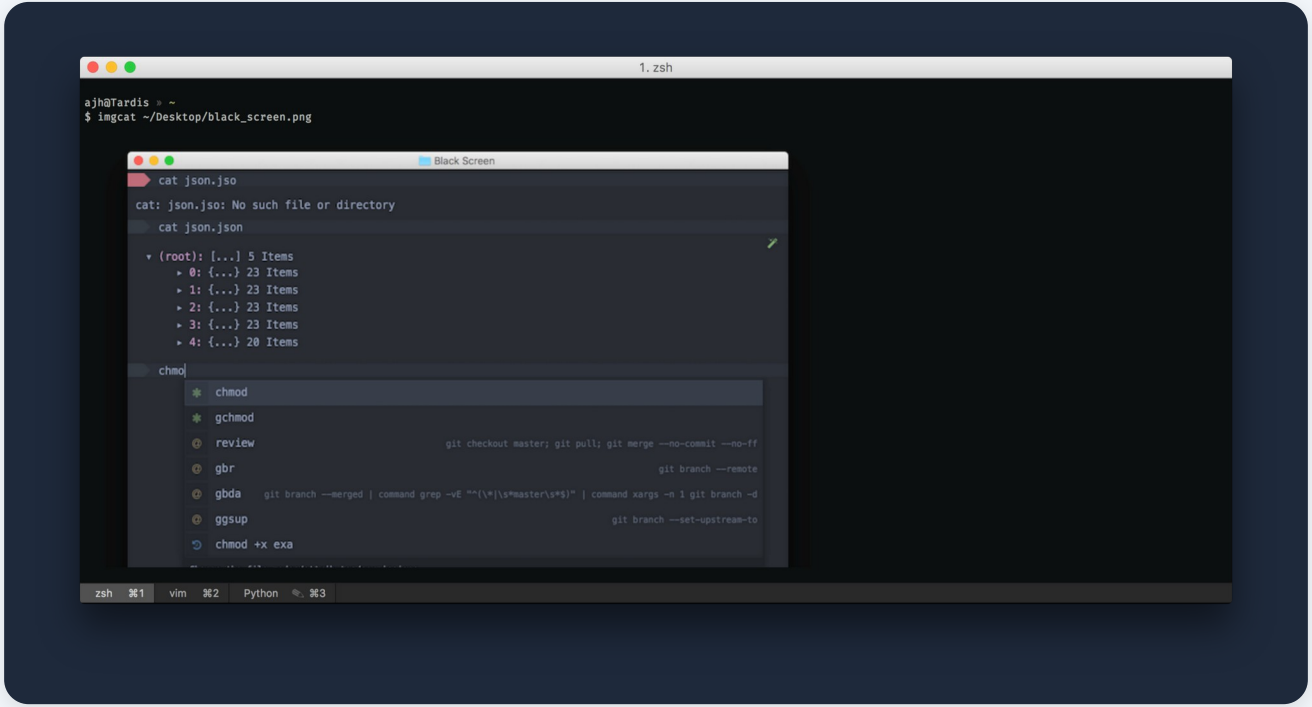
On desktop systems, the most popular user interfaces are the GUI shells.



Most popular interfaces are based on the X Window System, often simply called “X”.

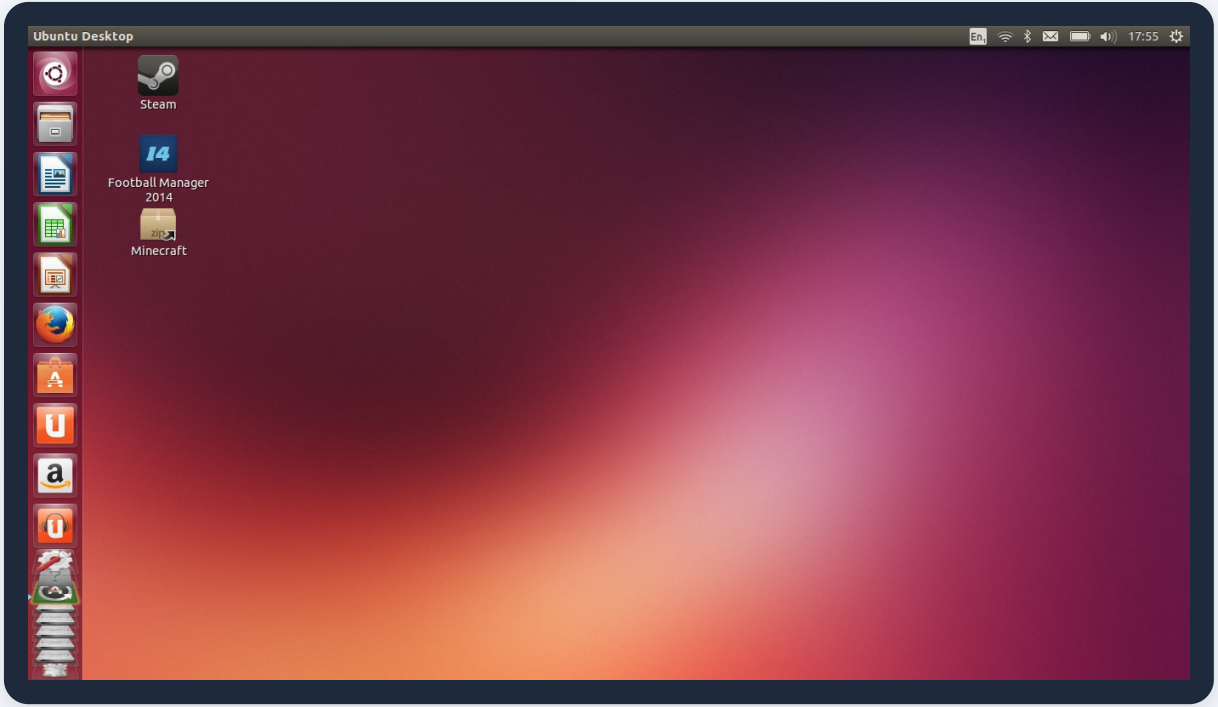
INTERFACE

CLI View of Linux



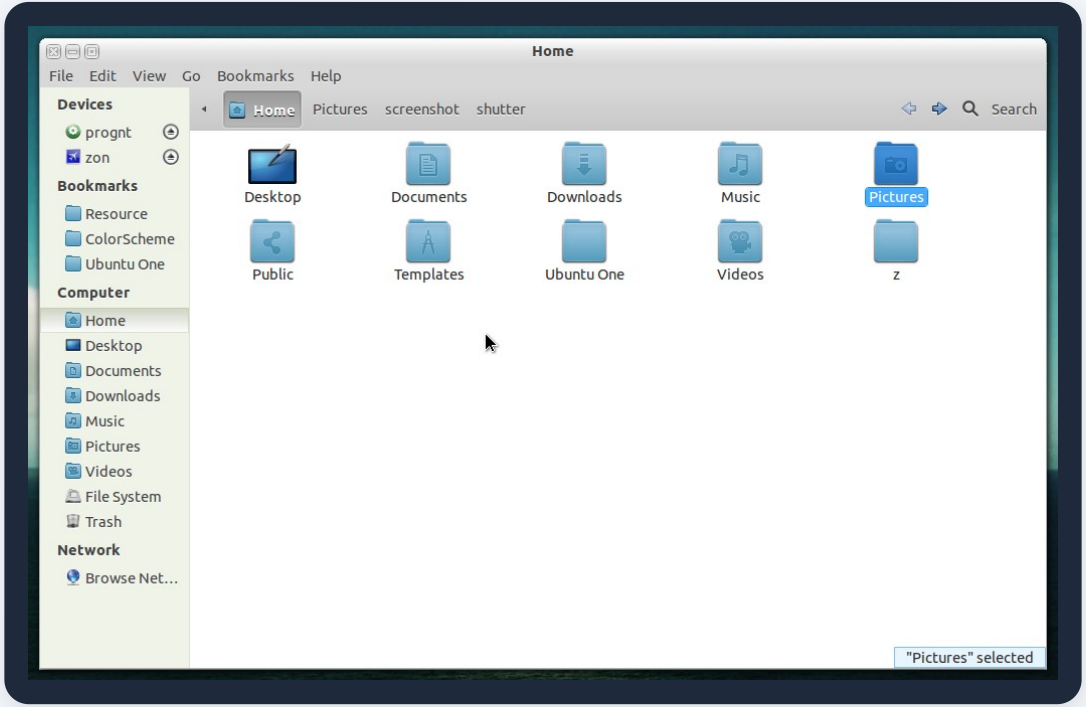
INTERFACE

Desktop View of Linux



INTERFACE

Inside View of Linux



ECOSYSTEM

Distributions of Linux

C

Corel Linux

D

Debian GNU/Linux

O

OpenLinux (Caldera)

R

Red Hat

U

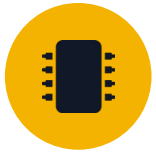
Ubuntu

T

TurboLinux

ECOSYSTEM

Hardware Requirements



CPU



Main memory



Optical drive



Graphic card



Hard drive



Sound card

ECOSYSTEM

Software Applications

OpenOffice

Word processing, spreadsheets and drawing.

Adobe Acrobat Reader

A viewer for PDF documents.

Konqueror

The KDE file manager and web browser.

TV, Video, Radio & Webcam

Multimedia applications.

ECOSYSTEM

Editors of Linux

>_ Vi / Vim

>_ Gedit

>_ Nano

>_ GNU Emacs

>_ Kate / Kwrite

>_ Lime Text

...and many more.

NEXT UP

Comparison with Other Operating Systems

```
$ diff linux windows ios
```

COMPARISON

Linux vs Windows

01 Licensing

Linux is freely available to download, while companies have to pay for a Windows licence.

02 Updates

Windows needs updating from time to time, and its update process is slower than Linux.

03 Compatibility

Linux supports backward compatibility, unlike Windows.

04 Software

Most Windows software must be licensed, but in Linux nearly all of it is freely available.

COMPARISON

Linux vs iOS

Aspect	iOS	Linux
Hardware requirement	Restrictive hardware requirements	No restrictive hardware requirements
Customizability	Restrictions on how data and display are arranged	Can be set up as simply as you want
Security	Both are highly secured and do not give direct permission to the system administrator.	

USAGE

Important Commands of Linux

Commands that give direct access to files through the terminal.

```
$ ls        # List command
$ mv        # Move command
$ mkdir     # Make directories
$ rmdir     # Remove directory
$ locate    # Locate directory
```

USAGE

Why Use Linux?



Costless



Stable



Reliable



Extremely powerful



Highly secure

USAGE

Merits and Demerits



Merits

- Easily accessible on old computers.
- Used for both commercial and personal purposes.



Demerits

- Not made for gaming.
- Not easy to understand for people new to Linux.
- Mostly used by programmers.
- For home use, Windows is mostly preferred.

USAGE

Use of Linux in Various Fields



Android app development



Routers and transmitting devices



Game designing

Defence and Education

Banking and Government sectors

USAGE

Commercial Use of Linux



From hobbyists to production

Adoption in production environments, beyond hobbyists, began in the mid-1990s with supercomputing.



Servers to supercomputers

Linux runs everything from embedded systems to supercomputers, and holds a secure place in server installations such as the popular LAMP stack.



Mobile with Android

Linux is also a major success in mobile development through Android, offering easy access and strong security.

Conclusion

“When is it best to use Linux and when should some other operating system be preferred?”

It all depends on the user.



Linux is only for programmer-based environments, not for non-programmers, because it is difficult to understand.

Thank You

